



Question Set

Question 1

A portfolio has two regimes. Calm regime has mean daily PnL 0.1 million and volatility 1 million. Stress regime has mean -2 million and volatility 5 million. Stress probability is 8 percent. Compute unconditional mean and variance.

Question 2

Prove the tower property

$$\mathbb{E}[\mathbb{E}[X | Y]] = \mathbb{E}[X]$$

Question 3

A loss model has

$$\mathbb{E}[L | S] = 3 + 4S$$

where S is a binary stress indicator with

$$\mathbb{P}(S = 1) = 0.1$$

Compute

$$\mathbb{E}[L]$$

Question 4

Show that if

$$X$$

is independent of

$$Y$$

then

$$\mathbb{E}[X | Y] = \mathbb{E}[X]$$

Question 5

A desk reports

$$\mathbb{E}[L | L > \text{VaR}_{99}]$$

as expected shortfall. What conditioning event is being used and what caveat applies for discrete losses?

Question 6

Derive the best linear predictor of Y from X in mean-square error form.

Question 7

A bond has loss given default LGD and default indicator D. If exposure E is known today, write $E[\text{loss} \mid \text{information today}]$.

Question 8

Explain conditional VaR forecasting in a volatility-regime model.

Question 9

Prove the orthogonality property

$$E[(X - E[X \mid \mathcal{G}])Z] = 0$$

for every

$\mathcal{G}$

-measurable

Z

Question 10

A model estimates $P(\text{default} \mid \text{downgrade})=12$ percent and $P(\text{downgrade})=5$ percent. What additional information is needed to compute $P(\text{downgrade} \mid \text{default})$?

Question 11

Why is $E[L \mid \text{stress}]$ not a stress loss scenario by itself?

Question 12

Use total expectation to decompose expected portfolio loss by rating class.